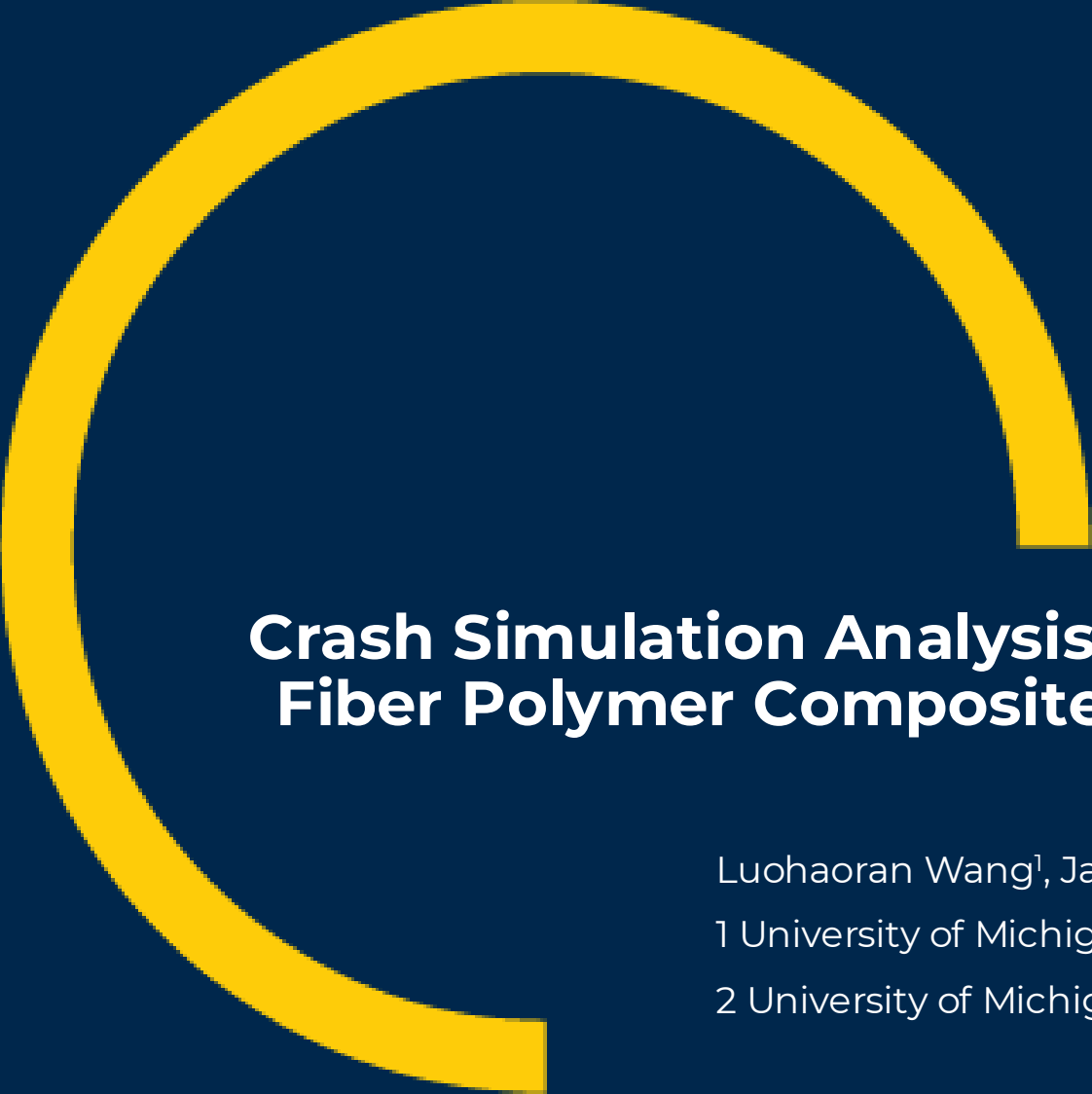




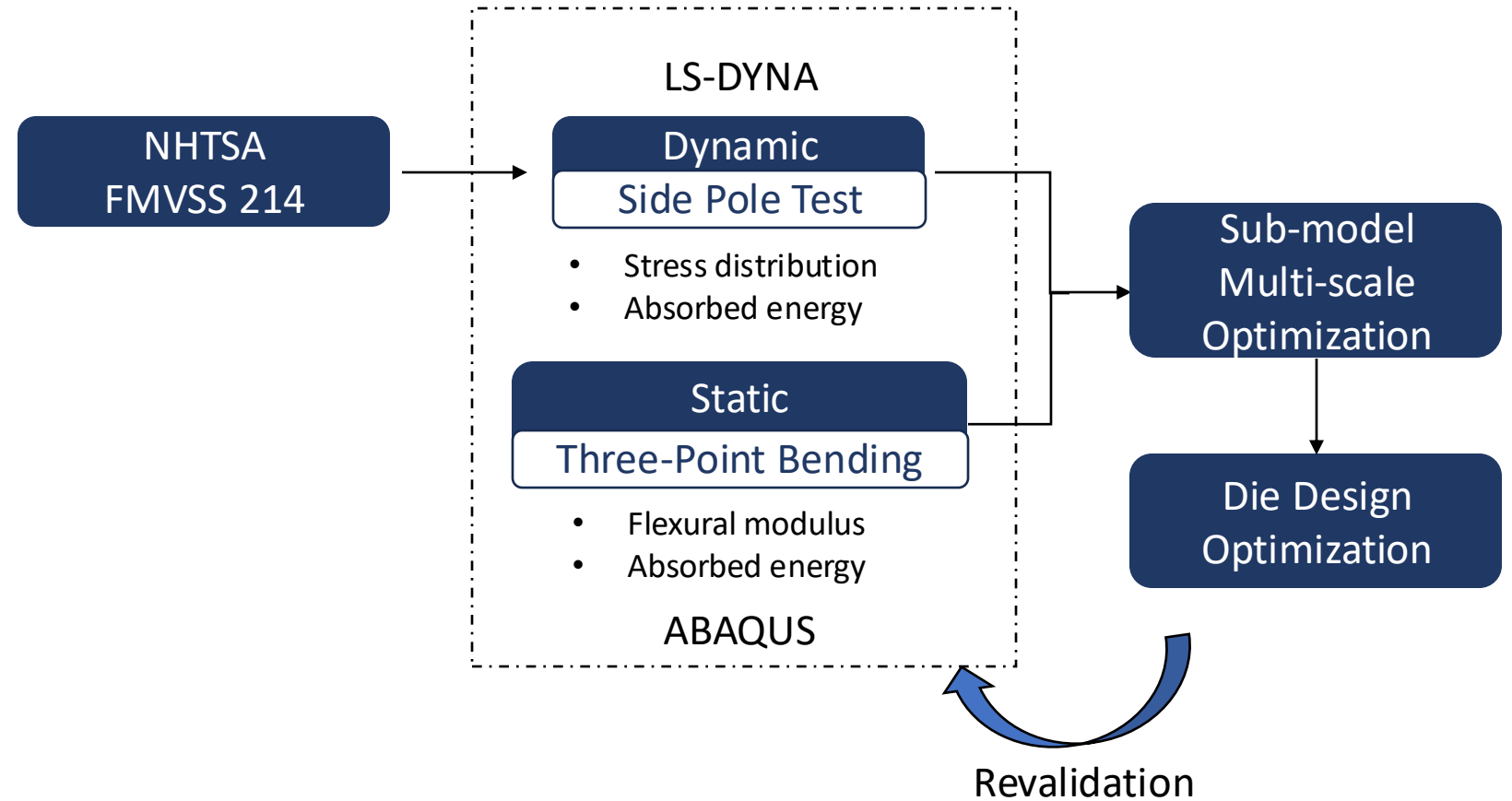
Crash Simulation Analysis for Optimization of the Carbon Fiber Polymer Composite Pultruded Battery Separators

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- Background & Introduction
- Modeling and Simulation
- Methodology
- Results & Discussion
- Conclusion



Background - Pultrusion

- Highest volume fraction of fiber
- Lowest cost compared to other manufacturing methods

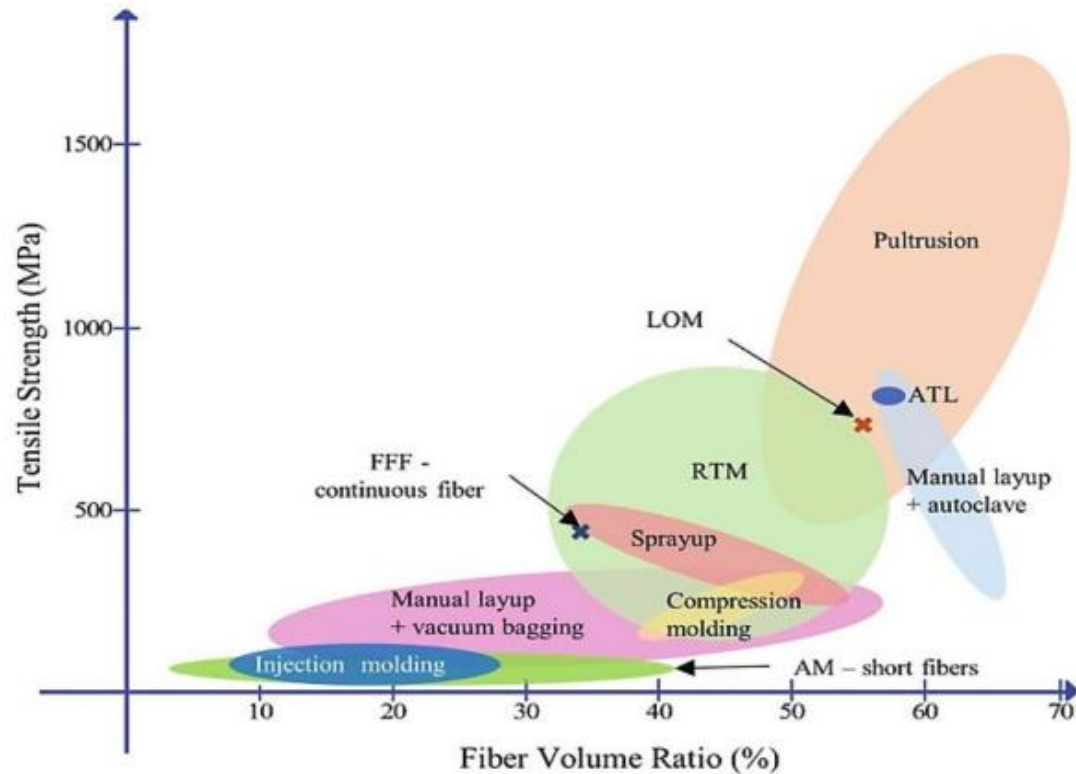


Figure 1. Tensile strength and FVF for manufacturing methods

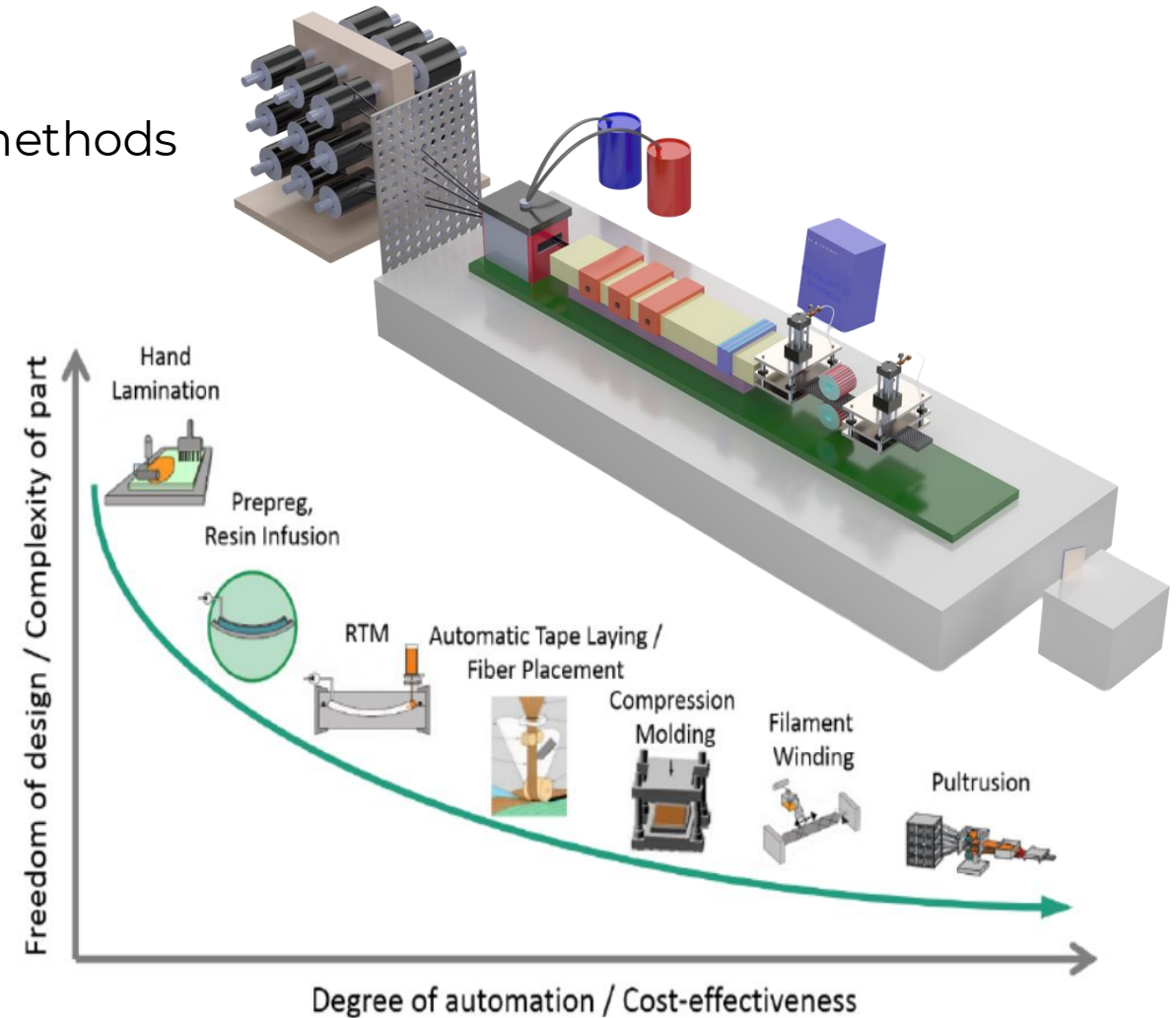


Figure 2. Cost effectiveness versus complexity of parts

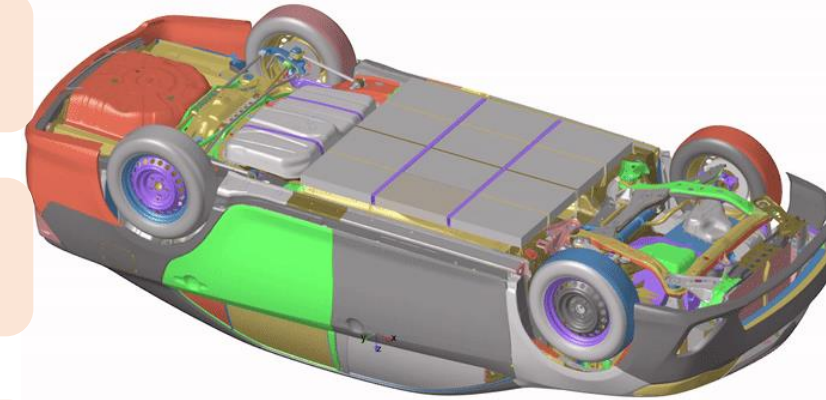
Battery Separators & Carbon fiber composites

Separator Functions

- Support cell modules under mechanical loading
- Dissipate heat through integration with cooling plates
- Isolate electrical and thermal zones between cells
- Enhance crash resistance without adding weight (*X. Zhang, 2023*)

Pultruded CFRP

- High stiffness-to-weight ratio
- Good energy absorption under bending & compression
- Thermal and dimensional stability under crash/heat
- Suited for continuous pultrusion manufacturing



Optimize battery separator

- Weight reduction
- High stiffness

Crash Simulation-Based Design

- Use rigid side-pole crash simulations to evaluate loading scenarios

- 50% increase in specific strength compared to steel
- 60% weight reduction vs. steel
- 25% reduction in greenhouse gas emission

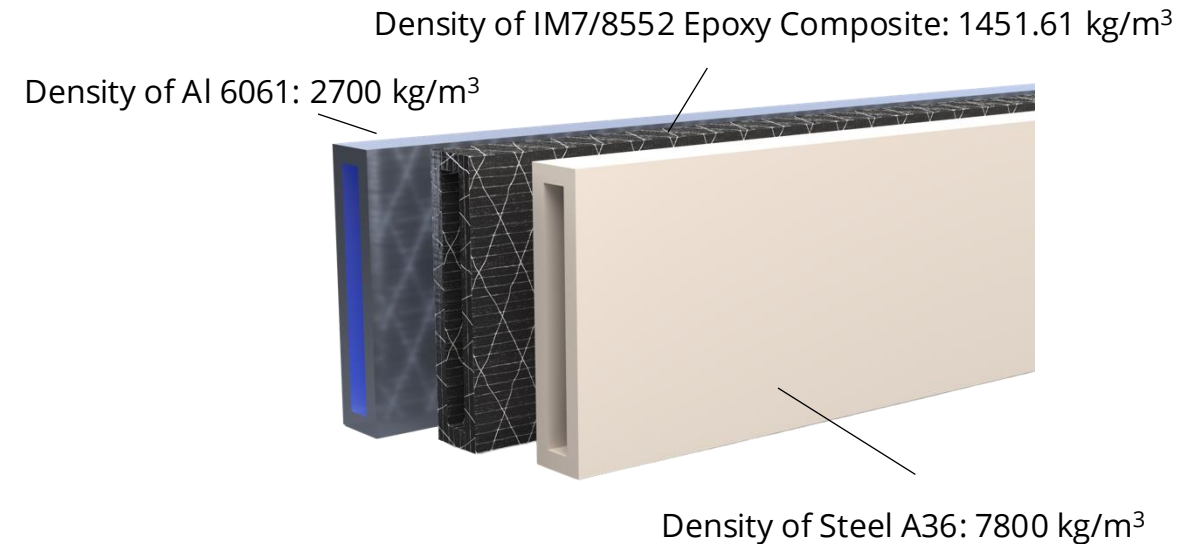
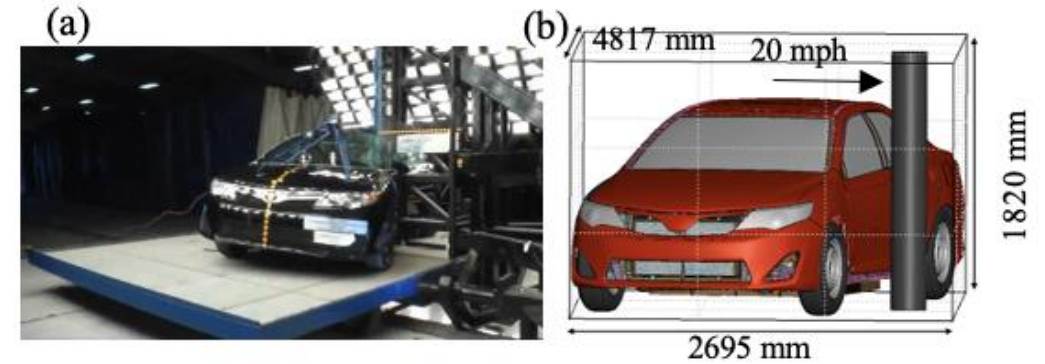


Figure 3. Material comparisons between aluminum, CFRP, and steel rendered by Solidworks 2024

Separator Geometry & Modeling

A 2015 Toyota Camry model (validated George Mason University) with embedded honeycomb reinforcement was investigated (Z. Mustaffa *et al.*, 2024).

Isomass T-shape and hollow rectangle shapes designed based on the design from BASF and embedded battery module.

Separator designs tailored to fit within the battery pack layout while enhancing mechanical integrity and crash protection.

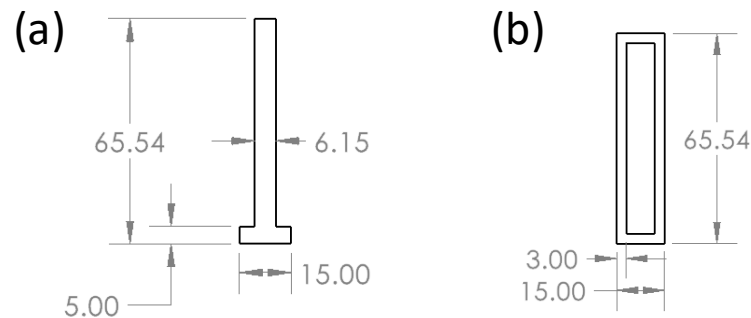


Figure 4. Designed separators (a) T-shaped and (b) hollow rectangle separators.

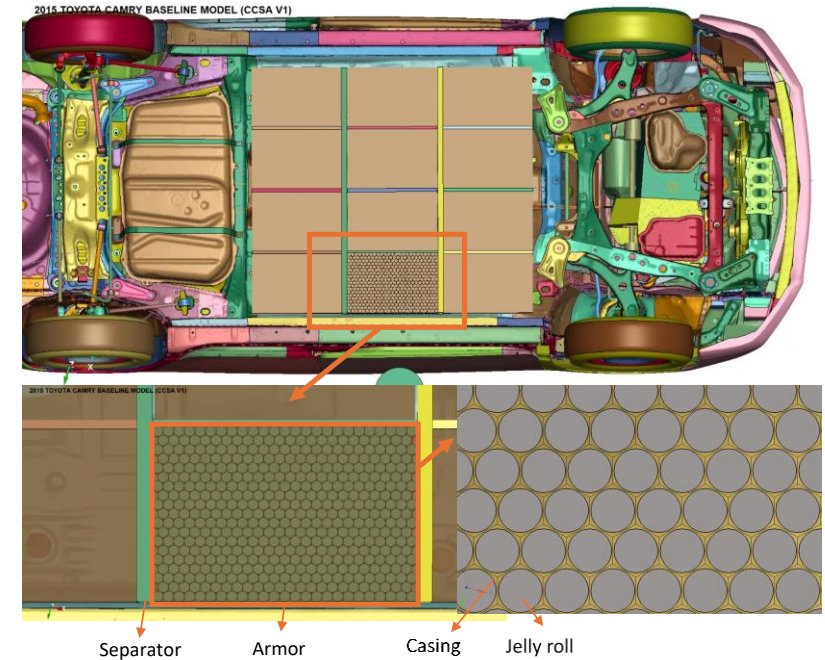


Figure 5. Whole vehicle model. (a) battery design with separators and a honeycomb structure and (b) honeycomb design and dimensions. (Z. Mustaffa *et al.*, 2024)

Material Properties for Honeycomb



Table 1. Mechanical properties of honeycomb reinforced component. (Z. Mustaffa et al., 2024)

Properties	Unit	Shell Casing and Cap	Jellyroll	Al 3003 Honeycomb Structure
Material name	-	MAT024. Piecewise Linear Plasticity	MAT063. Crushable Foam	MAT018. Power Law Plasticity
Mass density	kg/m ³	7850	12.75	2740
Elastic Modulus	GPa	200	0.5	68.95
Poisson's ratio	-	0.3	0.03	0.33
Yield Stress	GPa	0.342	-	-
Failure Strain	-	0.06	-	-

Material Properties for Composite

Table 2. Mechanical properties of IM7/8557 Composite. (Kaddour, A. S, 2013)

Property	Unit	Value
Mass density, ρ	kg/m ³	1580
Young's modulus; long. direction, E_A	GPa	165
Young's modulus – trans. direction, E_B	GPa	9
Poisson's ratio, $\nu_{12}=\nu_{13}$	-	0.34
Poisson's ratio, ν_{23}	-	0.5
Shear modulus, G_{AB} / G_{CA}	GPa	5.6
Shear modulus B_C , G_{BC}	GPa	2.8

NHTSA side pole test FMVSS 214 model (NHTSA.)

- A rigid pole was positioned diagonally at an angle of 75 ± 3 degrees in front of the A-pillar of a car.
- Vehicle model with the honeycomb battery module struck the pole at a speed of 20 mph within 10 seconds.
- The utility pole modeled as a rigid body constrained in six degrees of freedom.

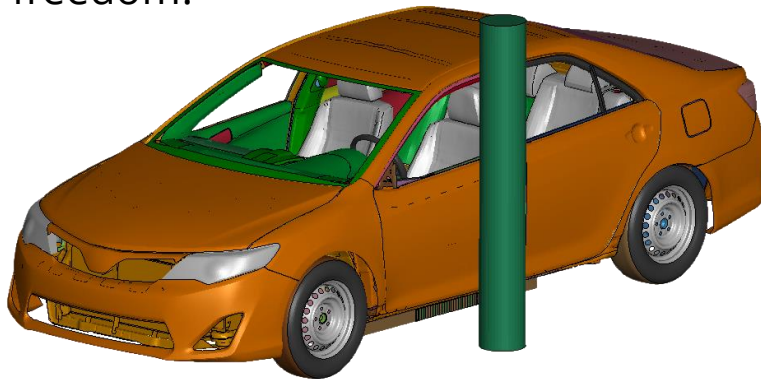


Figure 6. Side pole test in LS-DYNA 2024.

Customized three-point bending test

- Supporting cylinders positioned at 1/10 of total length on either side
- Velocity of 1mm/min assigned to the loading cylinders for the bending model within 200s
- Validate and optimize the mechanical properties of separators

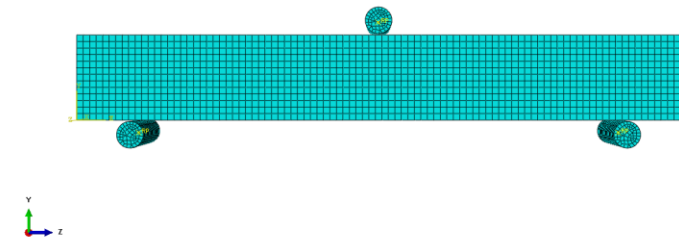


Figure 7. Schematic of the customized three-point bending test.

Methodology – Validation of Model

Table 3. Comparison of Center of Gravity Between Technical Specifications of Hybrid Toyota Camry and Simulation.

Axis	Estimated CG (from spec)	Simulated CG (from image)	Difference
X	2328 mm	2165.59 mm	162.41 mm rearward
Y	−80 mm	−79.31 mm	0.69 mm
Z	535 mm	496.94 mm	38.06 mm lower

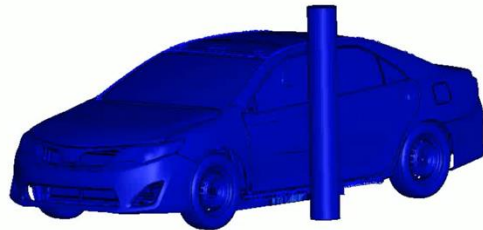


Figure 8. Comparison between EV model and Validated FEM Model (Courtesy of George Mason University)

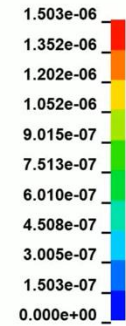
Results - Side Pole Test

(a)

2015 TOYOTA CAMRY BASELINE MODEL (CCSA V1)

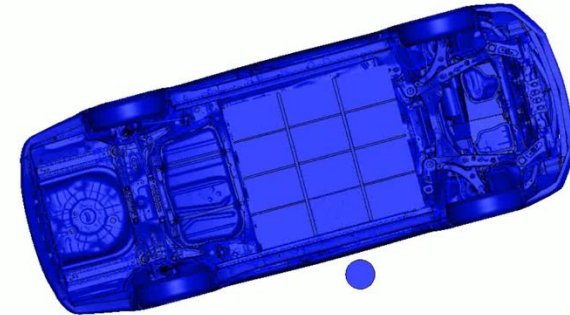


Von Mises Stress (MPa)

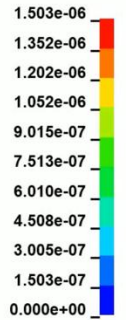


(b)

2015 TOYOTA CAMRY BASELINE MODEL (CCSA V1)



Von Mises Stress (MPa)



Animation 1. (a) Side view (b) Bottom view of side pole crash test in LS-DYNA 2024.

Results - Side Pole Test

Smallest displacements located at Figure. 8 (c) a

The most significant fracture failures occurred at the beam joint corners, where stress reached approximately **180 MPa**

Elevated stress concentrations at vertical separators due to point compression from neighboring horizontal separators

Impact damage occurring on the vertical beam's side caused by crashing of adjacent cell modules and horizontal separators.

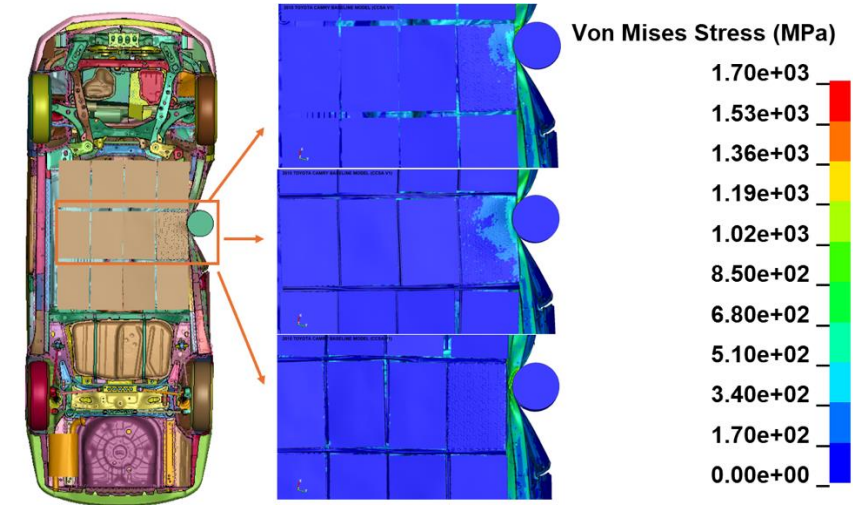


Figure 9. Side pole test results for EV (a) without separators, with (b) T-shaped, and (c) hollow rectangle separators.

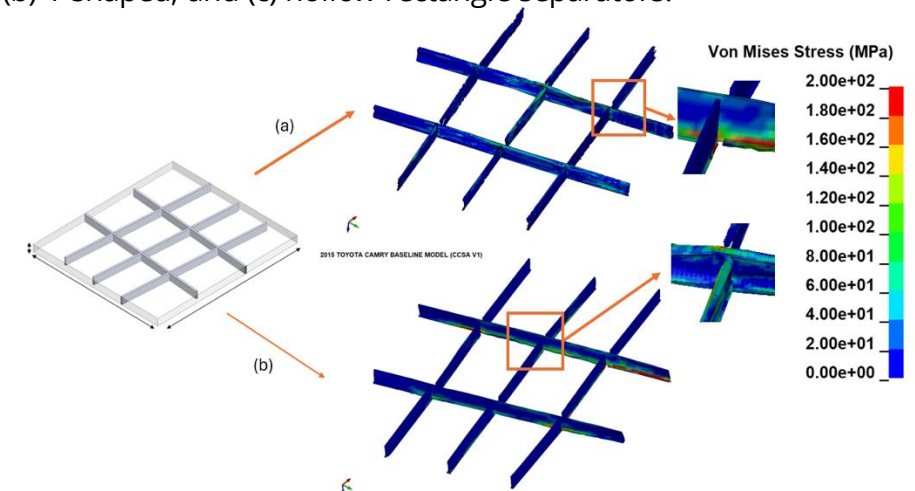


Figure 10. Von-Mises stress distribution for (a) T-shaped and (b) hollow rectangle separators.

Results - Side Pole Test

EV impact Energy Absorbed

- T-shape: **80.2 kJ**
- Hollow Rectangle: **87.4 kJ**
(highest absorption)

Embedded separators absorbed **0.12 kJ** and **0.35 kJ** of energy for the T-shaped and hollow rectangular separators

Initial beam failure appears around **0.01** seconds, energy absorbed until 0.03 seconds

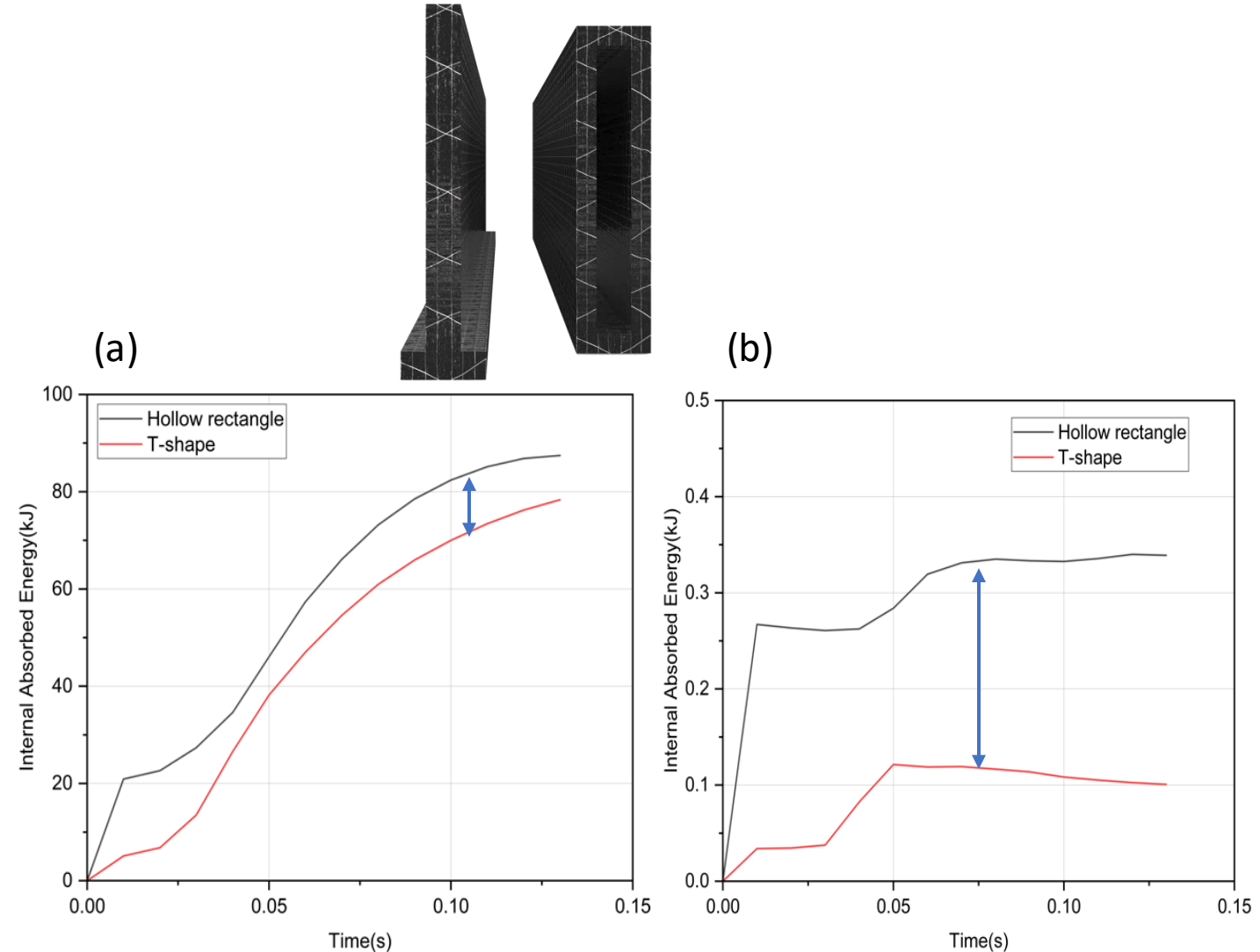


Figure 11. Total energy comparison; (a) full EV model; (b) battery separators.

Results - Three-point Bending Test

- **T-shape failed early** due to shear-induced failure
- **Lower bending moment** in T-shape is caused by asymmetric stiffness and smaller second moment of area, leading to buckling
- **Fluctuation** in the curve indicates damage onset or micro-cracking in CFRP
- T-shape has **8.46% lower stiffness** than the hollow rectangle, based on beam analysis

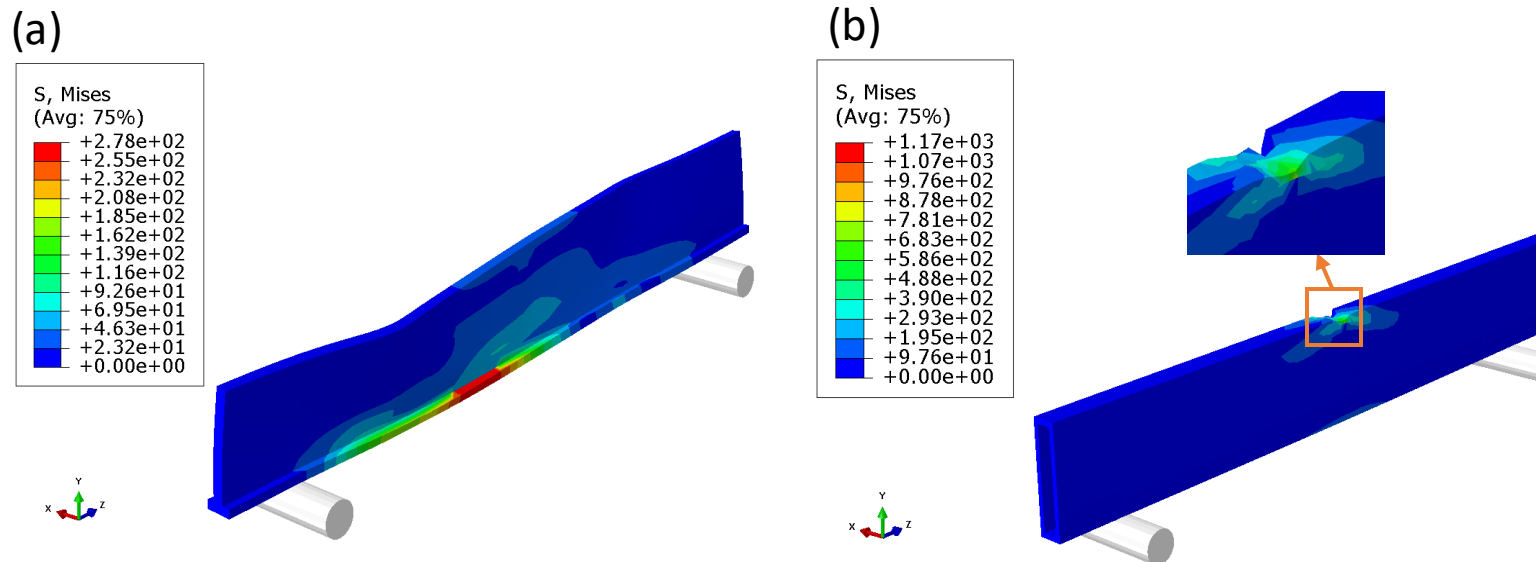


Figure 12. Stress comparison for (a) T-shaped and (b) hollow rectangle separators with CFRP.

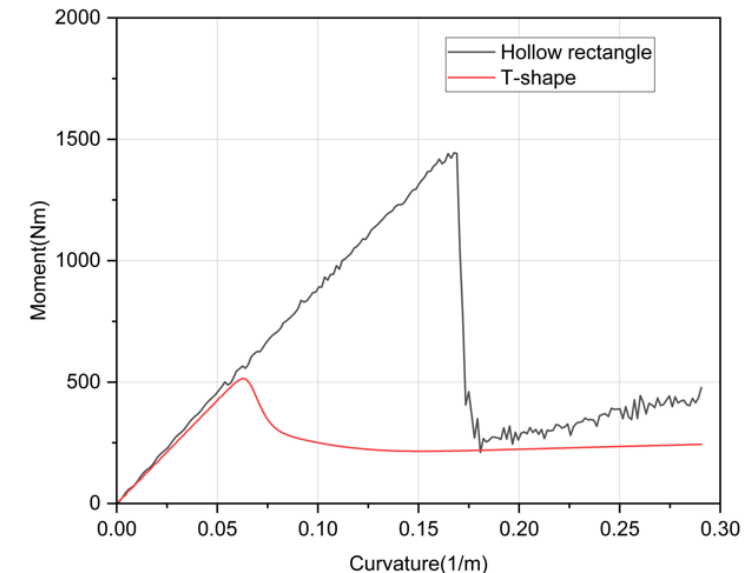


Figure 13. Moment vs curvature for T-shaped and hollow rectangle separators with CFRP.

Results - Three-point Bending Test

Wall thickness initiated at 3 – 7.5 mm (guided by EPTA) and increased in 0.5 mm increments.

As wall thickness increases, flexural modulus drops due to the faster decline in second moment ratio than the gain in axial stiffness. (H. M. Abdulhusain et al., 2020).

Bending stiffness is crucial for toughness and energy absorption, excess stiffness can reduce performance

Must balance stiffness, energy absorption, and geometry (O. S. David-West et al., 2008).

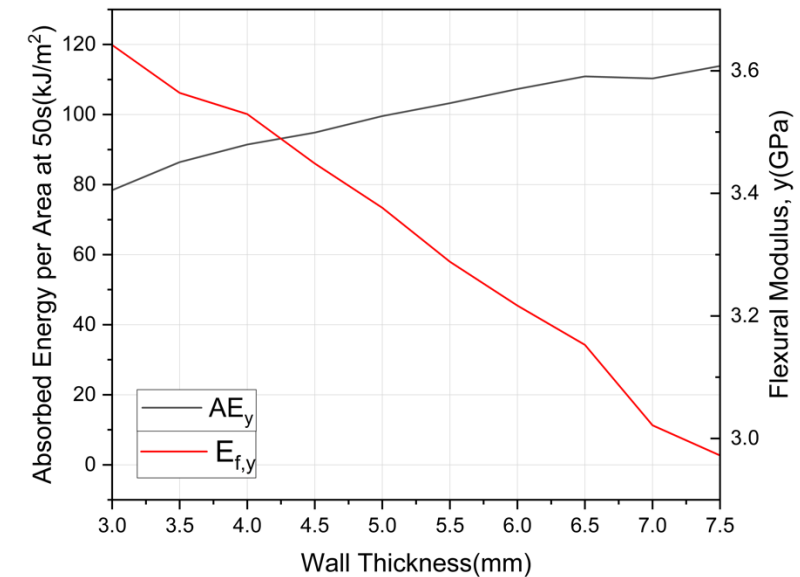
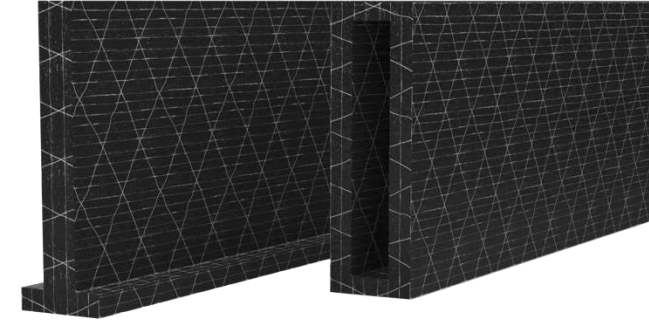


Figure 14. Three-point bending results for hollow rectangle separator with different wall thickness

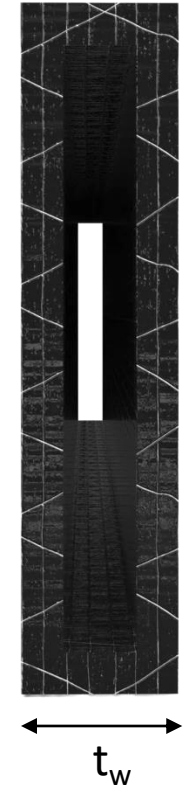
Results - Optimization of the Wall Thickness

Weight of the separator (W), was analyzed in relation to wall thickness

$$\mathbf{Max} E_{f,y}(t_w) = -0.1514t_w + 4.1162 \quad (1)$$

$$\mathbf{Max} AE_y(t_w) = 0.0381 \ln(t_w) + 0.0379 \quad (2)$$

$$\mathbf{Min} W(t_w) = [(1048.64) - (15 - 2t_w)(65.54 - 2t_w)] \quad (3)$$



Results - Optimization of the Wall Thickness

Regularization method was applied to introduce tunable parameters for optimal value determination (Mark J. L. Orr, 1995).

The weight function (F), is defined where w_1 (1.5), w_2 (1.5), and w_3 (0.5) are presented in the weighting ratio

$$F(t_w) = w_1 E_{f,y} + w_2 A E_y + \frac{w_3}{W} \quad (4)$$

The regularization term $-\lambda(5-t_w)^4$ was introduced to penalize deviations from the target thickness of 5 mm.

$$F(t_w) = w_1 E_{f,y} + w_2 A E_y + \frac{w_3}{W} - \lambda(5 - t_w)^4 \quad (5)$$

Optimal wall thickness = 3.98 mm (~4 mm)

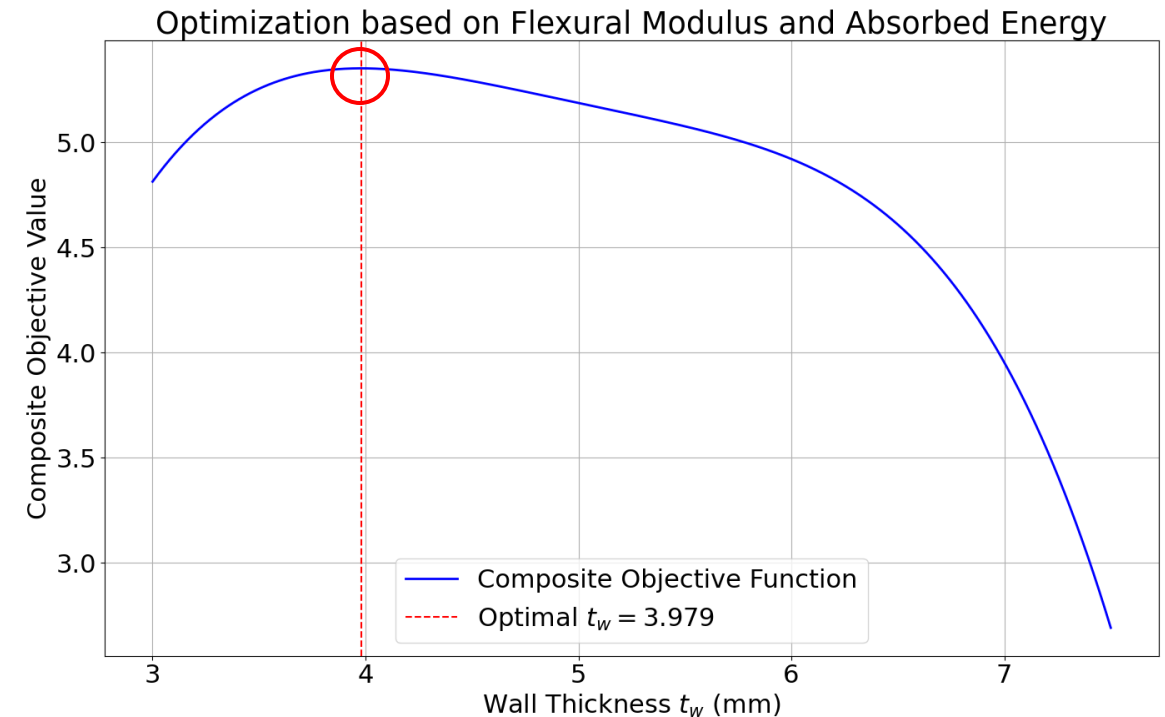


Figure 15. The optimized wall thickness of hollow rectangle separator with regularization selected center

Results – Optimum Comparison

The most significant fracture failures still occurred at the joint corners, where stress reached approximately **200 MPa**

The vertical separators compressed by neighboring horizontal separators and the horizontal separators were compressed by battery modules

Meanwhile, the 4 mm wall thickness hollow rectangle separators maintained their structure without complete deformation

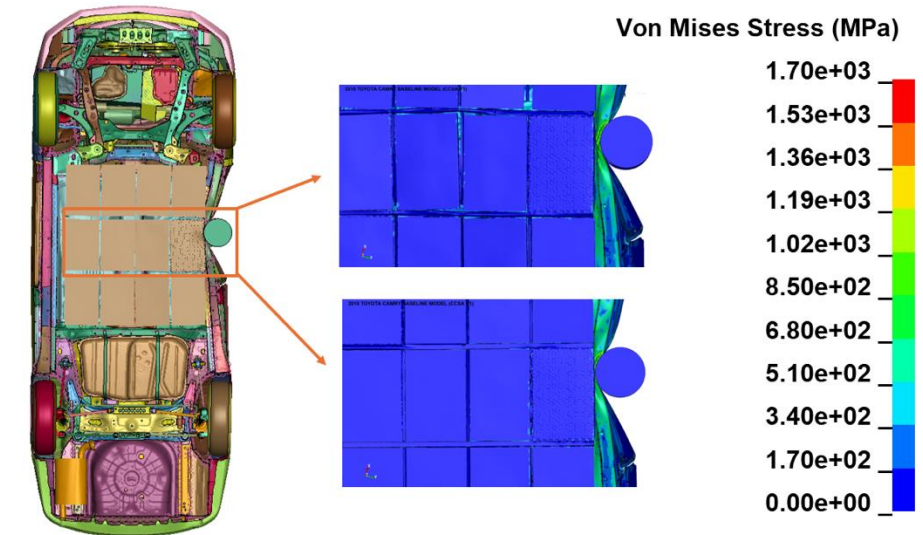


Figure 16. Side pole test results for EV with (a) 3mm, with (b) 4mm wall thickness hollow rectangle separators.

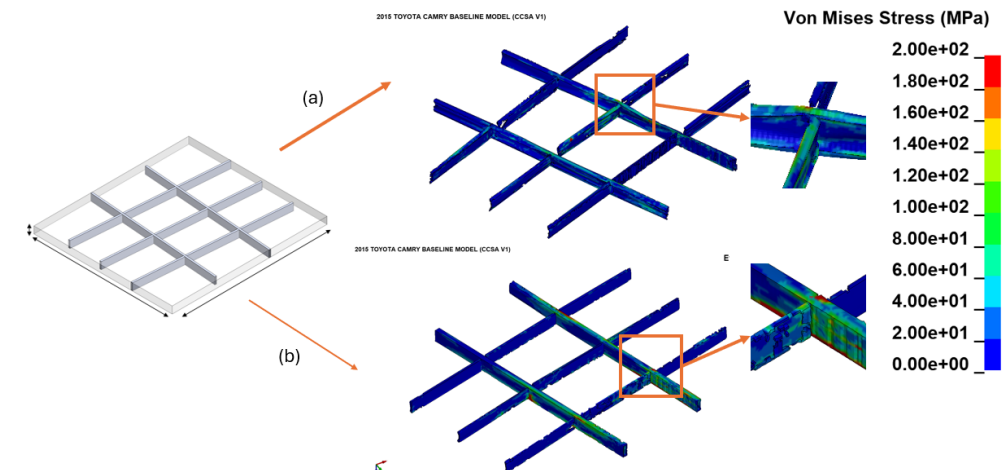
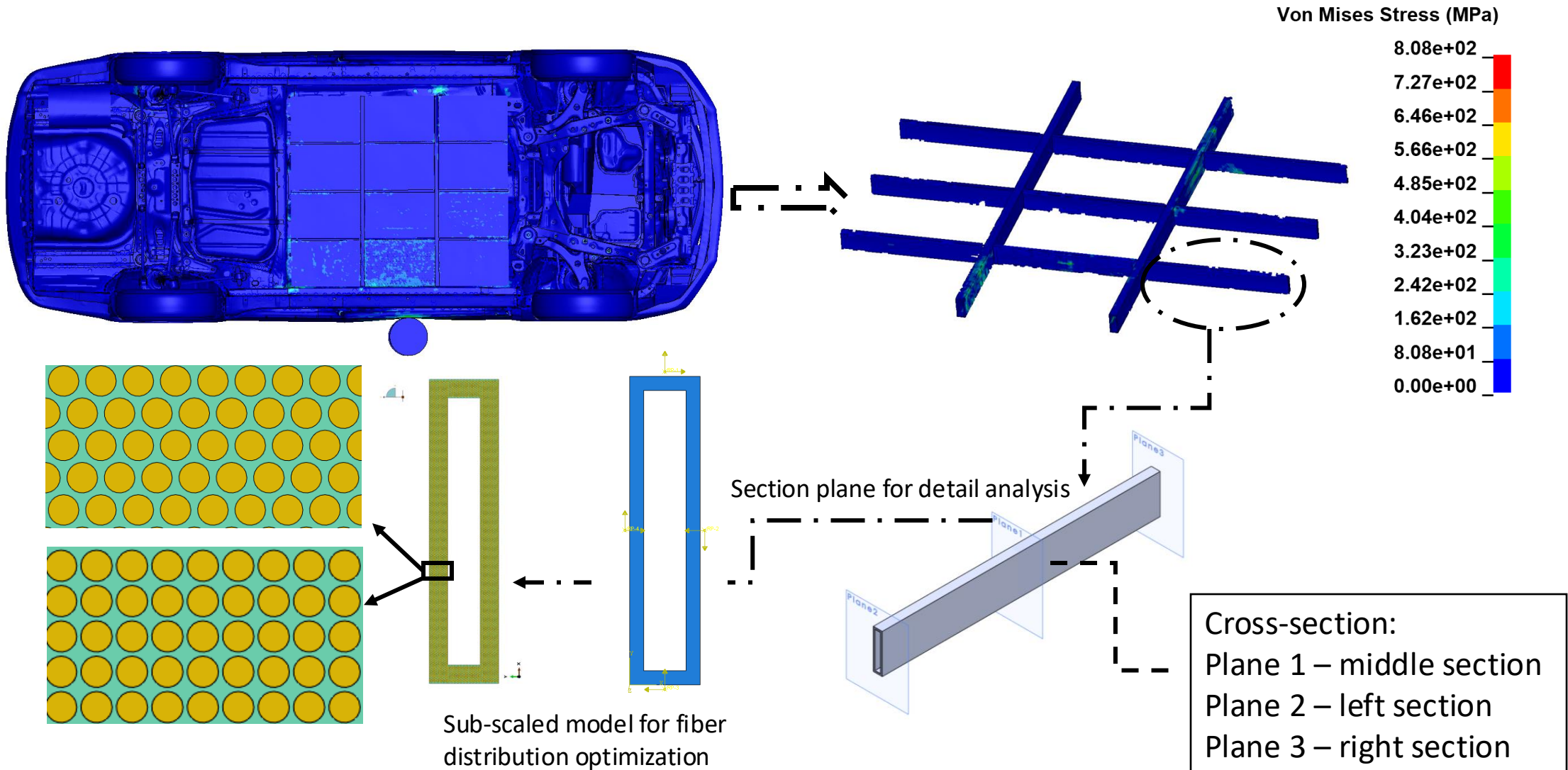


Figure 17. Von-Mises stress distribution for (a) 3mm and (b) 4mm wall thickness hollow rectangle separators.

Sub-model Multi-scale Optimization



- **CFRP separators achieved nearly 80% weight reduction** compared to conventional steel alloy components.
- **Side pole crash simulations accurately predicted loading behavior** of the separators under impact conditions.
- Pultruded CFRP separators enhanced energy absorption during impact.
- **At equal weight, hollow rectangular separator car** absorbed **7.2 kJ** more energy than T-shaped one.
- **Customized 3-point bending tests revealed earlier shear failure and 8.46% lower stiffness in T-shaped separators**
- **Using the regularization center method**, an optimal wall thickness of **4 mm was determined** for hollow rectangular separators

Acknowledgement



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Thank you!

Q&A